Before you start

Do these things first. Tick each box when it is done.

☐ **1. Set up the iPhone.**

- ☐ Settings > Camera > Formats > **Most Compatible**.
- ☐ Settings > Camera > **Level ON** (on an older iPhone: **Grid ON**).
- ☐ Use the **1x lens**. Do not use 0.5x or 2x.
- ☐ **Flash OFF**.

☐ **2. Clean white table.** In every photo there may ONLY be: the seeds, the ruler, the printed marker card and the sticky note. No feet, legs, hands, floor, phone, cup, pen or other objects.

☐ **3. Things you need.** Put them on the table now:

- ☐ Kitchen scale
- ☐ The steel ruler
- ☐ Sticky notes and a pen
- ☐ This checklist
- ☐ The empty bowl
- ☐ The printed marker card (cut out, with its white border)
- ☐ The bucket with seeds
- ☐ A pen to fill in this checklist

☐ **4. Write these sticky notes now**, one name on each note, in big letters:
P01, P02, P03, P03-R, P03-T, P04, P05, P06, P07, P08, P08-R, P08-T, P09, P10, P11, P12, P13, P13-R, P13-T, P14, P15

- ☐ **21 notes written.**

☐ **5. Marker check.** Put the ruler on the printed 100 mm line of the marker sheet.

Line length: _____ mm (must be 100 mm)

☐ **6. Start.** Write the time and the light.

Start time: _____

Light (circle one): **daylight** / **room light**

Rules that always apply

- Press **TARE** right before every weighing.
- No seed may touch another seed (except in the T photos).
- Camera straight down, lens **40 cm** above the table. Wait until the crosshair in the camera turns **yellow**.
- Everything must be inside the photo: all seeds, ruler, marker card and note. And nothing else.
- Never clean the seeds. Broken hull pieces stay in.

PILE P01: target 10 g

Sticky note name: **P01** | Target weight: **10 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

☐ **1.** Stir the seeds in the bucket with your hand.

☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.

☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **10 g** (1 g more or less is fine).

☐ **5.** Write down the weight of the seeds in the bowl:

Weight before: _____ g

☐ **6.** Pour all seeds from the bowl onto the white table.

☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.

☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.

☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.

☐ **10.** Put the sticky note **P01** at the side, at least **3 cm** away from the seeds.

☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.

☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P01 are in the picture, and nothing else.

☐ **13.** Take **PHOTO A** of pile P01.

Photo A taken: ☐

☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.

☐ **15.** Take **PHOTO B** of pile P01.

Photo B taken: ☐

PILE P01: target 10 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P01 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P01).

☐ **22.** Count the photos you took for pile P01. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P02: target 12 g

Sticky note name: **P02** | Target weight: **12 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

☐ **1.** Stir the seeds in the bucket with your hand.

☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.

☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **12 g** (1 g more or less is fine).

☐ **5.** Write down the weight of the seeds in the bowl:

Weight before: _____ g

☐ **6.** Pour all seeds from the bowl onto the white table.

☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.

☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.

☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.

☐ **10.** Put the sticky note **P02** at the side, at least **3 cm** away from the seeds.

☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.

☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P02 are in the picture, and nothing else.

☐ **13.** Take **PHOTO A** of pile P02.

Photo A taken: ☐

☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.

☐ **15.** Take **PHOTO B** of pile P02.

Photo B taken: ☐

PILE P02: target 12 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P02 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P02).

☐ **22.** Count the photos you took for pile P02. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P03: target 14 g

Sticky note name: **P03** | Target weight: **14 g** | Photos for this pile: **4**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

☐ **1.** Stir the seeds in the bucket with your hand.

☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.

☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **14 g** (1 g more or less is fine).

☐ **5.** Write down the weight of the seeds in the bowl:

Weight before: _____ g

☐ **6.** Pour all seeds from the bowl onto the white table.

☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.

☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.

☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.

☐ **10.** Put the sticky note **P03** at the side, at least **3 cm** away from the seeds.

☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.

☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P03 are in the picture, and nothing else.

☐ **13.** Take **PHOTO A** of pile P03.

Photo A taken: ☐

☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.

☐ **15.** Take **PHOTO B** of pile P03.

Photo B taken: ☐

PILE P03: target 14 g (continued)

Part 2: Extra photos (only for pile P03)

☐ **16.** Sweep all seeds together with your hand. Spread them again in a new way, so that no seed touches another seed.

☐ **17.** Replace the sticky note **P03** with the sticky note **P03-R**.

☐ **18.** Hold the phone straight down at 40 cm, crosshair yellow. Check: all seeds, ruler, marker card and note P03-R are in the picture, and nothing else. Take the **R PHOTO**.

R photo taken: ☐

☐ **19.** Push all seeds together with your hand so that they touch each other. No seed may lie on top of another seed.

☐ **20.** Replace the sticky note **P03-R** with the sticky note **P03-T**.

☐ **21.** Hold the phone straight down at 40 cm, crosshair yellow. Check: all seeds, ruler, marker card and note P03-T are in the picture, and nothing else. Take the **T PHOTO**.

T photo taken: ☐

PILE P03: target 14 g (continued)

Part 3: Weigh again and put the seeds back

☐ **22.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **23.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **24.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **25.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **26.** Pour the seeds from the bowl back into the bucket.

☐ **27.** Does pile P03 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P03).

☐ **28.** Count the photos you took for pile P03. It should be 4. (This pile has the extra R and T photos.)

Number of photos taken for this pile: _____ (should be 4)

☐ **29.** Write the time.

Time finished: _____

☐ **30.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P04: target 16 g

Sticky note name: **P04** | Target weight: **16 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

☐ **1.** Stir the seeds in the bucket with your hand.

☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.

☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **16 g** (1 g more or less is fine).

☐ **5.** Write down the weight of the seeds in the bowl:

Weight before: _____ g

☐ **6.** Pour all seeds from the bowl onto the white table.

☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.

☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.

☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.

☐ **10.** Put the sticky note **P04** at the side, at least **3 cm** away from the seeds.

☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.

☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P04 are in the picture, and nothing else.

☐ **13.** Take **PHOTO A** of pile P04.

Photo A taken: ☐

☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.

☐ **15.** Take **PHOTO B** of pile P04.

Photo B taken: ☐

PILE P04: target 16 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P04 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P04).

☐ **22.** Count the photos you took for pile P04. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P05: target 18 g

Sticky note name: **P05** | Target weight: **18 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **18 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P05** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P05 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P05.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P05.

  Photo B taken: ☐

PILE P05: target 18 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P05 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P05).

☐ **22.** Count the photos you took for pile P05. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P06: target 20 g

Sticky note name: **P06** | Target weight: **20 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **20 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P06** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P06 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P06.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P06.

  Photo B taken: ☐

PILE P06: target 20 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P06 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P06).

☐ **22.** Count the photos you took for pile P06. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P07: target 22 g

Sticky note name: **P07** | Target weight: **22 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **22 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P07** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P07 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P07.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P07.

  Photo B taken: ☐

PILE P07: target 22 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P07 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P07).

☐ **22.** Count the photos you took for pile P07. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P08: target 24 g

Sticky note name: **P08** | Target weight: **24 g** | Photos for this pile: **4**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

☐ **1.** Stir the seeds in the bucket with your hand.

☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.

☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **24 g** (1 g more or less is fine).

☐ **5.** Write down the weight of the seeds in the bowl:

Weight before: _____ g

☐ **6.** Pour all seeds from the bowl onto the white table.

☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.

☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.

☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.

☐ **10.** Put the sticky note **P08** at the side, at least **3 cm** away from the seeds.

☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.

☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P08 are in the picture, and nothing else.

☐ **13.** Take **PHOTO A** of pile P08.

Photo A taken: ☐

☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.

☐ **15.** Take **PHOTO B** of pile P08.

Photo B taken: ☐

PILE P08: target 24 g (continued)

Part 2: Extra photos (only for pile P08)

☐ **16.** Sweep all seeds together with your hand. Spread them again in a new way, so that no seed touches another seed.

☐ **17.** Replace the sticky note **P08** with the sticky note **P08-R**.

☐ **18.** Hold the phone straight down at 40 cm, crosshair yellow. Check: all seeds, ruler, marker card and note P08-R are in the picture, and nothing else. Take the **R PHOTO**.

R photo taken: ☐

☐ **19.** Push all seeds together with your hand so that they touch each other. No seed may lie on top of another seed.

☐ **20.** Replace the sticky note **P08-R** with the sticky note **P08-T**.

☐ **21.** Hold the phone straight down at 40 cm, crosshair yellow. Check: all seeds, ruler, marker card and note P08-T are in the picture, and nothing else. Take the **T PHOTO**.

T photo taken: ☐

PILE P08: target 24 g (continued)

Part 3: Weigh again and put the seeds back

☐ **22.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **23.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **24.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **25.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **26.** Pour the seeds from the bowl back into the bucket.

☐ **27.** Does pile P08 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P08).

☐ **28.** Count the photos you took for pile P08. It should be 4. (This pile has the extra R and T photos.)

Number of photos taken for this pile: _____ (should be 4)

☐ **29.** Write the time.

Time finished: _____

☐ **30.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P09: target 26 g

Sticky note name: **P09** | Target weight: **26 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **26 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P09** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P09 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P09.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P09.

  Photo B taken: ☐

PILE P09: target 26 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P09 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P09).

☐ **22.** Count the photos you took for pile P09. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P10: target 28 g

Sticky note name: **P10** | Target weight: **28 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **28 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P10** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P10 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P10.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P10.

  Photo B taken: ☐

PILE P10: target 28 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P10 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P10).

☐ **22.** Count the photos you took for pile P10. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P11: target 30 g

Sticky note name: **P11** | Target weight: **30 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **30 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P11** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P11 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P11.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P11.

  Photo B taken: ☐

PILE P11: target 30 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P11 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P11).

☐ **22.** Count the photos you took for pile P11. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P12: target 32 g

Sticky note name: **P12** | Target weight: **32 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **32 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P12** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P12 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P12.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P12.

  Photo B taken: ☐

PILE P12: target 32 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P12 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P12).

☐ **22.** Count the photos you took for pile P12. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P13: target 34 g

Sticky note name: **P13** | Target weight: **34 g** | Photos for this pile: **4**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **34 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P13** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P13 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P13.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P13.

  Photo B taken: ☐

PILE P13: target 34 g (continued)

Part 2: Extra photos (only for pile P13)

☐ **16.** Sweep all seeds together with your hand. Spread them again in a new way, so that no seed touches another seed.

☐ **17.** Replace the sticky note **P13** with the sticky note **P13-R**.

☐ **18.** Hold the phone straight down at 40 cm, crosshair yellow. Check: all seeds, ruler, marker card and note P13-R are in the picture, and nothing else. Take the **R PHOTO**.

R photo taken: ☐

☐ **19.** Push all seeds together with your hand so that they touch each other. No seed may lie on top of another seed.

☐ **20.** Replace the sticky note **P13-R** with the sticky note **P13-T**.

☐ **21.** Hold the phone straight down at 40 cm, crosshair yellow. Check: all seeds, ruler, marker card and note P13-T are in the picture, and nothing else. Take the **T PHOTO**.

T photo taken: ☐

PILE P13: target 34 g (continued)

Part 3: Weigh again and put the seeds back

☐ **22.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **23.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **24.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **25.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **26.** Pour the seeds from the bowl back into the bucket.

☐ **27.** Does pile P13 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P13).

☐ **28.** Count the photos you took for pile P13. It should be 4. (This pile has the extra R and T photos.)

Number of photos taken for this pile: _____ (should be 4)

☐ **29.** Write the time.

Time finished: _____

☐ **30.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P14: target 36 g

Sticky note name: **P14** | Target weight: **36 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **36 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P14** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P14 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P14.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P14.

  Photo B taken: ☐

PILE P14: target 36 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P14 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P14).

☐ **22.** Count the photos you took for pile P14. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

PILE P15: target 38 g

Sticky note name: **P15** | Target weight: **38 g** | Photos for this pile: **2**

Remember: press TARE before every weighing. No seed touches another seed. Phone straight down at 40 cm. Wait for the yellow crosshair.

Part 1: Weigh the seeds and take the photos

- ☐ **1.** Stir the seeds in the bucket with your hand.
- ☐ **2.** Nothing on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **3.** Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.
- ☐ **4.** Scoop seeds from the bucket into the bowl until the scale shows about **38 g** (1 g more or less is fine).
- ☐ **5.** Write down the weight of the seeds in the bowl:

  Weight before: _____ g
- ☐ **6.** Pour all seeds from the bowl onto the white table.
- ☐ **7.** Spread the seeds with your fingers so that **NO seed touches another seed**. Keep them in one compact area.
- ☐ **8.** Put the ruler flat along the bottom side of the seeds, at least **3 cm** away from the seeds.
- ☐ **9.** Put the marker card flat in a corner, at least **3 cm** away from the seeds.
- ☐ **10.** Put the sticky note **P15** at the side, at least **3 cm** away from the seeds.
- ☐ **11.** Stand the ruler upright next to the seeds. Hold the phone so the back camera lens is at the **40 cm** mark. Then lay the ruler flat again in its place.
- ☐ **12.** Hold the phone straight down at 40 cm. Wait until the crosshair turns **yellow**. Check: all seeds, ruler, marker card and note P15 are in the picture, and nothing else.
- ☐ **13.** Take **PHOTO A** of pile P15.

  Photo A taken: ☐
- ☐ **14.** Move the phone away from the table. Bring it back to 40 cm, straight down, crosshair yellow.
- ☐ **15.** Take **PHOTO B** of pile P15.

  Photo B taken: ☐

PILE P15: target 38 g (continued)

Part 2: Weigh again and put the seeds back

☐ **16.** Take the empty bowl. Nothing on the scale: press **TARE**. Put the empty bowl on the scale. Press **TARE**. The scale shows **0 g**.

☐ **17.** Put **ALL** seeds from the table back into the bowl. Check the table: no seed left.

☐ **18.** Write down the weight of the seeds in the bowl:

Weight after: _____ g

☐ **19.** If "before" and "after" differ by more than 1 g, write what happened:

_____

☐ **20.** Pour the seeds from the bowl back into the bucket.

☐ **21.** Does pile P15 fit in the photo with no seeds touching?

Circle one: yes / no
If NO: stop here, do not do the next piles, and send a message with the number of this pile (P15).

☐ **22.** Count the photos you took for pile P15. It should be 2.

Number of photos taken for this pile: _____ (should be 2)

☐ **23.** Write the time.

Time finished: _____

☐ **24.** Remarks (light changed, seed fell, anything strange):

_____

_____

Upload the photos

Do this after all piles are done. Use your phone.

☐ **1.** Open the **Google Drive** app.

☐ **2.** Go to: **My Drive > KU Leuven AI 2025-28 > 2026-27_Master > S1_Applied-AI-Industrial_B30776 > 02_CNH_Data-Collection > 03_Assignment > photos**

☐ **3.** Tap **+** > **New folder**. Name the folder: **main**

☐ **4.** Open the folder **main**.

☐ **5.** Tap **+** > **Upload**. Choose **all photos of today**.

☐ **6.** Count the photos in the folder **main**.

Count of photos uploaded: _____ (should be 36 if all 15 piles fit)

☐ **7.** Take a photo of **every page** of this filled-in checklist and send it.

Summary table

Copy the numbers from the pile pages into this table.

Pile	Target (g)	Before (g)	After (g)	Photos	Fits? (yes/no)
P01	10				
P02	12				
P03	14				
P04	16				
P05	18				
P06	20				
P07	22				
P08	24				
P09	26				
P10	28				
P11	30				
P12	32				
P13	34				
P14	36				
P15	38				

Total photos: _____ (should be 36 if all 15 piles fit)

Monday 19 October

Bring these things

- [ ] The jar with **150 g** of seeds for Jia
- [ ] A small bag with about **15 g** of seeds for the test
- [ ] The kitchen scale
- [ ] The ruler
- [ ] The marker card
- [ ] Sticky notes
- [ ] The laptop

Meeting

- [ ] **15:00 to 15:30:** online meeting with Jia.

Monday 19 October: small test with Jia

In class, during the project time (from about 19:30)

- [ ] **1.** Write a sticky note: **GRR**.
- [ ] **2.** Put a white sheet of paper on the table.
- [ ] **3.** Spread the **15 g** of seeds on the white sheet. **No seed touches another seed.**
- [ ] **4.** Put the marker card and the note **GRR** next to the seeds. Nothing else in the picture.
- [ ] **5.** Hadi takes a photo with **Hadi's phone**.
- [ ] **6.** Hadi takes a photo with **Jia's phone**.
- [ ] **7.** Jia takes a photo with **Jia's phone**.
- [ ] **8.** Jia takes a photo with **Hadi's phone**.
- [ ] **9.** Write which photo is which:

  Photo 1: taken by _____ with phone of _____

  Photo 2: taken by _____ with phone of _____

  Photo 3: taken by _____ with phone of _____

  Photo 4: taken by _____ with phone of _____

- [ ] **10.** Weigh the seeds.

  Hadi weighs the seeds: _____ g

  Jia weighs the seeds: _____ g

- [ ] **11.** Ask CNH the 5 questions from STATUS_AND_PLAN section 7.